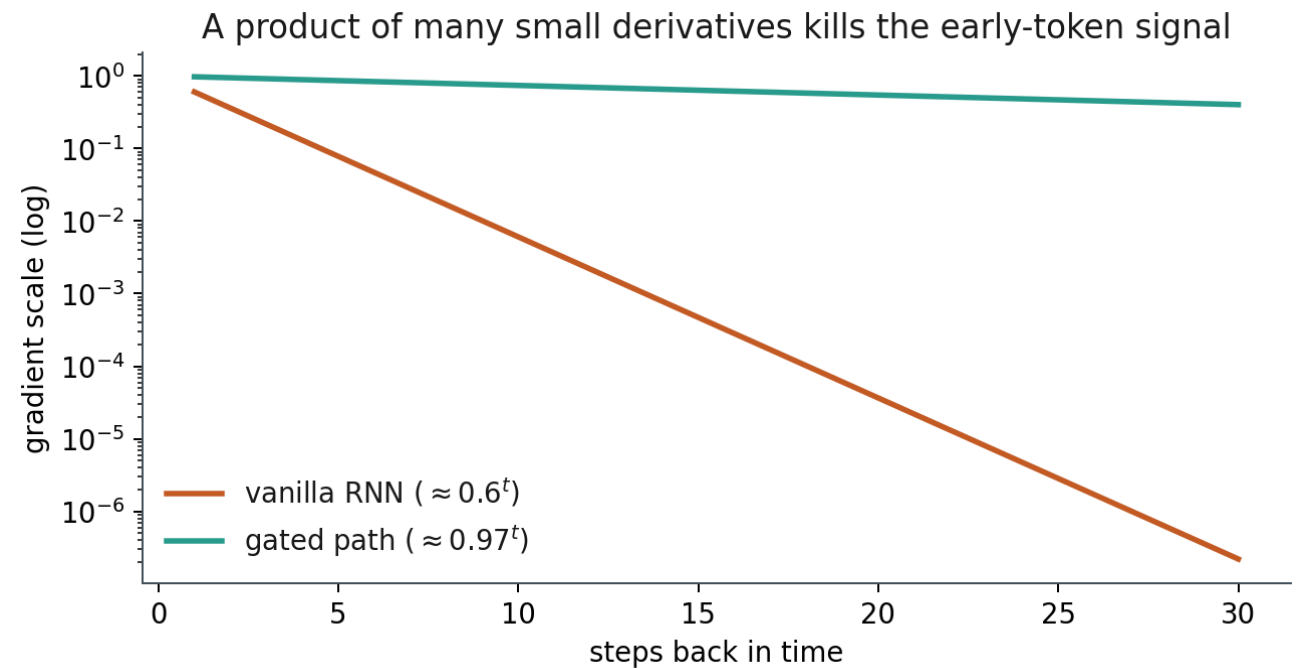


A product of many Jacobians

- $\partial L / \partial h_1$ multiplies one factor per step.
- $\tanh'$ is at most 1, and usually much smaller.
- Gates keep a path whose multiplier can stay near 1.



Two failure modes

- Vanishing: early tokens do not train.
- Exploding: NaNs. Clipping helps explosions, not vanishing.

